Unit 1: Whole Numbers

Ordinal Numbers

Ordinal numbers tell the position of things in a row: 1st (first), 2nd (second), 3rd (third), 4th (fourth), 5th (fifth), 6th (sixth), 7th (seventh), 8th (eighth), 9th (ninth), 10th (tenth), 11th (eleventh), 12th (twelfth), 13th (thirteenth), 14th (fourteenth), 15th (fifteenth), 16th (sixteenth), 17th (seventeenth), 18th (eighteenth), 19th (nineteenth), 20th (twentieth).

Exercise 1 — Alphabet Positions

A = 1st, B = 2nd, C = 3rd, D = 4th, E = 5th, F = 6th, G = 7th, H = 8th, I = 9th, J = 10th, K = 11th, L = 12th, M = 13th, N = 14th, O = 15th, P = 16th, Q = 17th, R = 18th, S = 19th, T = 20th.

Counting 1 to 100 in Words (blanks filled)

Page 4: the missing numerals are 11, 13, 19, 26; the missing words are 20 = Twenty, 28 = Twenty-eight.

Page 5: the missing numerals are 31, 38, 40, 47, 50, 59; the missing words are 43 = Forty-three, 44 = Forty-four, 48 = Forty-eight, 49 = Forty-nine, 58 = Fifty-eight.

Page 6: the missing numerals are 63, 66, 72, 73, 79, 85, 90; the missing words are 62 = Sixty-two, 74 = Seventy-four, 77 = Seventy-seven, 80 = Eighty.

Page 7: the missing numerals are 95, 98; the missing words are 99 = Ninety-nine, 100 = One Hundred.

Key Facts: 0 is the smallest 1-digit number. 9 is the greatest 1-digit number. 10 is the smallest 2-digit number. 99 is the greatest 2-digit number.

Try Yourself (page 7): 24, 46, 62, 79, 95.

Three-Digit Numbers

10 ones = 1 ten. 10 tens = 100 ones = 1 hundred. 100 is the smallest 3-digit number. Hundreds run from 100 to 900. Block readings: 126 = one hundred twenty-six; 342 = three hundred forty-two; 999 = nine hundred ninety-nine, the greatest 3-digit number.

Exercise 2 — Words, Numerals and Sequences

Words to numerals: Fifteen = 15 (given); Twenty-two = 22; Thirty-six = 36; Forty-one = 41; Fifty-eight = 58; Sixty-three = 63; Seventy = 70; Eighty-four = 84; Ninety-six = 96.

Numerals to words: 18 = Eighteen (given); 21 = Twenty-one; 33 = Thirty-three; 45 = Forty-five; 54 = Fifty-four; 69 = Sixty-nine; 76 = Seventy-six; 80 = Eighty; 99 = Ninety-nine.

Words to numerals (3-digit): one hundred fifty-two = 152 (given); three hundred thirty-eight = 338; four hundred fifty = 450; five hundred nine = 509; six hundred fifty-eight = 658; seven hundred eleven = 711; eight hundred sixty-eight = 868; nine hundred ninety-nine = 999.

Sequences: 115, 116, 117, 118, 119, 120, 121; 238, 239, 240, 241, 242, 243, 244; 582, 583, 584, 585, 586, 587, 588; 697, 698, 699, 700, 701, 702, 703; 877, 878, 879, 880, 881, 882, 883.

Numerals given: 152, 338, 450, 509, 658, 711, 868, 999.

Numerals in words: 275 = two hundred seventy-five; 432 = four hundred thirty-two; 560 = five hundred sixty; 689 = six hundred eighty-nine; 345 = three hundred forty-five; 709 = seven hundred nine; 811 = eight hundred eleven; 990 = nine hundred ninety.

Place Value

26 = 20 + 6 (2 tens and 6 ones). 245 = 200 + 40 + 5 (2 hundreds, 4 tens, 5 ones). 308 = 3 hundreds, 0 tens, 8 ones. 400 = 4 hundreds, 0 tens, 0 ones. 333 = 300 + 30 + 3 (Try Yourself).

Value of the coloured digit: in 472 the 4 = 400; in 238 the 3 = 30.

Exercise 3 — Place Value

Splits: 136 = 100 + 30 + 6; 285 = 200 + 80 + 5; 412 = 400 + 10 + 2; 360 = 300 + 60 + 0; 507 = 500 + 0 + 7; 771 = 700 + 70 + 1; 649 = 600 + 40 + 9; 800 = 800 + 0 + 0; 999 = 900 + 90 + 9.

Coloured-digit values: 125 -> 20; 270 -> 200; 598 -> 8; 418 -> 400; 600 -> 600; 301 -> 0; 764 -> 700; 996 (middle 9) -> 90; 850 -> 0.

Sums: 132, 215, 550, 402, 719, 888, 600, 996.

Place-value builds: 251, 534, 660, 507, 198, 300.

Comparison and Ordering

435 > 85, because a 3-digit number is always greater than a 2-digit number. 518 > 376 (the book prints 375 once, a misprint for 376). 368 > 321. 469 > 463. Rule: when the digits are equal in number, compare the hundreds first, then the tens, then the ones.

Ordering example: 235, 516, 147. Ascending: 147, 235, 516. Descending: 516, 235, 147. Also 162, 203, 168. Ascending: 162, 168, 203. Descending: 203, 168, 162.

Exercise 4 — Greater, Smaller, Ascending, Descending

Greater number: 121, 248, 218, 600, 749, 999. Smaller number: 89, 169, 289, 505, 699, 989.

Greatest of three: 135, 275, 390, 700, 809, 999. Smallest of three: 98, 183, 482, 142, 660, 889.

Ascending: (a) 105, 115, 125, 130, 135. (b) 435, 452, 498, 517, 579. (c) 809, 890, 895, 960, 990.

Descending: (a) 210, 205, 200, 198, 128. (b) 708, 650, 606, 599, 485. (c) 950, 850, 750, 650, 550.

Counting in 10s and 100s

Backward 10 steps from 112: 111, 110, 109, 108, 107, 106, 105, 104, 103, 102. From 343: 342, 341, 340, 339, 338, 337, 336, 335, 334, 333. Try Yourself from 96: 95, 94, 93, 92, 91, 90, 89, 88, 87, 86. From 125: 124, 123, 122, 121, 120, 119, 118, 117, 116, 115. From 278: 277, 276, 275, 274, 273, 272, 271, 270, 269, 268. From 350: 349, 348, 347, 346, 345, 344, 343, 342, 341, 340.

Counting in 10s: 40, 50, 60, 70, 80, 90, 100; 110, 120, 130, 140, 150, 160, 170; 360, 370, 380, 390, 400, 410, 420. Next six from 90: 100, 110, 120, 130, 140, 150; from 220: 230, 240, 250, 260, 270, 280; from 580: 590, 600, 610, 620, 630, 640; from 690: 700, 710, 720, 730, 740, 750.

Counting in 100s: 300, 400, 500, 600, 700, 800; 210, 310, 410, 510, 610, 710; 450, 550, 650, 750, 850, 950. Next five from 105: 205, 305, 405, 505, 605; from 330: 430, 530, 630, 730, 830; from 190: 290, 390, 490, 590, 690; from 444: 544, 644, 744, 844, 944.

One Thousand

999 + 1 = 1000. One thousand is the first and the smallest 4-digit number: 1 thousand, 0 hundreds, 0 tens, 0 ones.

I have learnt to: read and write ordinal numbers; count in words and numerals; use place value; compare and order numbers; count in 10s and

100s; recognise one thousand.

Vocabulary: Ordinal Numbers, Compare, Ascending Order, Descending Order.

Review Exercise (pages 25-26)

Multiple-choice answers: (i) In words, 46 is written as forty-six (b). (ii) Ordinal numbers are used to represent the position (c) of the objects. (iii) Nine hundred nine is written in numerals as 909 (d). (iv) In 158, the place value of 1 is 100 (c). (v) Of 989, 990, 909 and 999, the greatest number is 999 (a).

Alphabet positions: D = 4th, G = 7th, J = 10th, M = 13th, I = 9th, T = 20th, Q = 17th, K = 11th, S = 19th, N = 14th.

Words to numerals: 89, 313, 506, 786, 859, 976. Numerals to words: 96 = ninety-six; 269 = two hundred sixty-nine; 404 = four hundred four; 890 = eight hundred ninety; 967 = nine hundred sixty-seven.

Coloured-digit values: 589 -> 80; 490 -> 400; 756 -> 700; 600 -> 600; 850 -> 800; 915 -> 900.

10 steps down from 205: 204, 203, 202, 201, 200, 199, 198, 197, 196, 195.

Counting in 10s: 180, 190, 200, 210, 220, 230, 240, 250. Counting in 100s: 130, 230, 330, 430, 530, 630, 730.

Ascending: 105, 145, 405, 415, 501, 514. Descending: 514, 501, 415, 405, 145, 105.

Unit 2: Number Operations — Addition

Addition with Carrying (2-digit)

Opener: 5 + 3 = 8 balloons. 8 + 5 = 13: ones 8 + 5 = 13, write 3 and carry 1 ten; tens 0 + 1 = 1. Answer 13.

25 + 7 = 32. Ones: 5 + 7 = 12, write 2 and carry 1 ten. Tens: 2 + 1 = 3. Answer 32.

47 + 26 = 73. Ones: 7 + 6 = 13, write 3 and carry 1 ten. Tens: 4 + 2 + 1 = 7. Answer 73.

Key Fact: zero added to a number gives the number itself (for example 25 + 0 = 25).

Exercise 1 — Addition with Carrying

17 + 5 = 22; 35 + 6 = 41; 48 + 7 = 55; 55 + 8 = 63; 89 + 3 = 92; 25 + 15 = 40; 43 + 29 = 72; 56 + 35 = 91; 68 + 25 = 93; 12 + 79 = 91; 28 + 54 = 82; 67 + 26 = 93.

Amna has 24 books and Hina has 8 books. How many books do they have in all?

Step 1: 24 + 8. Step 2: ones 4 + 8 = 12, write 2 and carry 1 ten. Step 3: tens 2 + 1 = 3. Answer: 32 books.

Raza's class has 35 students and Maryam's class has 28 students. How many students are there in both classes?

Step 1: 35 + 28. Step 2: ones 5 + 8 = 13, write 3 and carry 1 ten. Step 3: tens 3 + 2 + 1 = 6. Answer: 63 students.

A fruit seller sold 36 kinnos in the morning and 48 kinnos in the evening. How many kinnos did he sell in all?

Step 1: 36 + 48. Step 2: ones 6 + 8 = 14, write 4 and carry 1 ten. Step 3: tens 3 + 4 + 1 = 8. Answer: 84 kinnos.

Three-Digit Addition without Carrying

132 + 6 = 138: ones 2 + 6 = 8; tens 3; hundreds 1. 316 + 82 = 398: ones 6 + 2 = 8; tens 1 + 8 = 9; hundreds 3. 435 + 362 = 797: ones 5 + 2 = 7; tens 3 + 6 = 9; hundreds 4 + 3 = 7.

Exercise 2 — Three-Digit Addition without Carrying

252 + 6 = 258; 165 + 3 = 168; 568 + 1 = 569; 680 + 6 = 686; 345 + 34 = 379; 426 + 70 = 496; 441 + 58 = 499; 607 + 82 = 689; 272 + 127 = 399; 562 + 437 = 999; 608 + 291 = 899; 242 + 544 = 786.

Rehan bought a bat for Rs 390 and a ball for Rs 208. How much did he spend in all?

Step 1: 390 + 208. Step 2: ones 0 + 8 = 8. Step 3: tens 9 + 0 = 9. Step 4: hundreds 3 + 2 = 5. Answer: Rs 598.

Three-Digit Addition with Carrying

School: 198 students + 6 new students = 204. Ones: 8 + 6 = 14, write 4 and carry 1 ten. Tens: 9 + 1 = 10, write 0 and carry 1 hundred. Hundreds: 1 + 1 = 2. Answer 204.

Garden: 254 mango trees + 78 guava trees = 332. Ones: 4 + 8 = 12, write 2 and carry 1 ten. Tens: 5 + 7 + 1 = 13, write 3 and carry 1 hundred. Hundreds: 2 + 1 = 3. Answer 332.

Farm: 345 cows + 265 sheep = 610. Ones: 5 + 5 = 10, write 0 and carry 1 ten. Tens: 4 + 6 + 1 = 11, write 1 and carry 1 hundred. Hundreds: 3 + 2 + 1 = 6. Answer 610.

Exercise 3 — Three-Digit Addition with Carrying

185 + 6 = 191; 249 + 6 = 255; 574 + 7 = 581; 795 + 8 = 803; 897 + 3 = 900; 533 + 49 = 582; 354 + 68 = 422; 209 + 91 = 300; 827 + 76 = 903; 739 + 62 = 801; 223 + 158 = 381; 384 + 126 = 510; 493 + 309 = 802; 616 + 288 = 904; 395 + 298 = 693.

Pakistan scored 462 runs in the first innings and 378 runs in the second innings. How many runs did Pakistan score in both innings?

Step 1: 462 + 378. Step 2: ones 2 + 8 = 10, write 0 and carry 1 ten. Step 3: tens 6 + 7 + 1 = 14, write 4 and carry 1 hundred. Step 4: hundreds 4 + 3 + 1 = 8. Answer: 840 runs.

Mental Addition Strategy

20 + 15 = 35: split 15 into 10 + 5; 20 + 10 = 30; 30 + 5 = 35. 32 + 17 = 49: split 17 into 10 + 7; 32 + 10 = 42; 42 + 7 = 49.

Practice: (a) 20 + 18 = 38. (b) 15 + 12 = 27. (c) 30 + 10 = 40. (d) 20 + 10 = 30. (e) 30 + 20 = 50. (f) 10 + 12 = 22. (g) 20 + 11 = 31. (h) 14 + 12 = 26. (i) 23 + 15 = 38. (j) 35 + 13 = 48. (k) 42 + 7 = 49.

Review — Addition

Multiple-choice answers: 25 + 0 = 25 (c). 100 + 10 = 110 (d). First add the ones (a). Zero added to a number gives the number itself (b). Finally add the hundreds (c).

Addition drill: (a) 9 + 6 = 15. (b) 16 + 8 = 24. (c) 59 + 9 = 68. (d) 78 + 2 = 80. (e) 37 + 25 = 62. (f) 51 + 39 = 90. (g) 64 + 16 = 80. (h) 73 + 18 = 91. (i) 254 + 5 = 259. (j) 323 + 6 = 329. (k) 609 + 30 = 639. (l) 812 + 77 = 889. (m) 312 + 185 = 497. (n) 602 + 296 = 898. (o) 457 + 8 = 465. (p) 694 + 7 = 701. (q) 357 + 56 = 413. (r) 589 + 88 = 677. (s) 798 + 133 = 931. (t) 555 + 298 = 853. (u) 666 + 195 = 861. (v) 599 + 308 = 907. (w) 777 + 169 = 946. (x) 696 + 294 = 990. (y) 459 + 367 = 826. (z) 888 + 77 = 965.

Ahmed bought a car for Rs 245 and a ball for Rs 75. How much did he spend?

Step 1: 245 + 75. Step 2: ones 5 + 5 = 10, write 0 and carry 1 ten. Step 3: tens 4 + 7 + 1 = 12, write 2 and carry 1 hundred. Step 4: hundreds 2 + 1 = 3. Answer: Rs 320.

Amna bought a teddy bear for Rs 384 and a ball for Rs 75. How much did she

spend?

Step 1: 384 + 75. Step 2: ones 4 + 5 = 9. Step 3: tens 8 + 7 = 15, write 5 and carry 1 hundred. Step 4: hundreds 3 + 1 = 4. Answer: Rs 459.

Irfan bought a bicycle for Rs 196 and a car for Rs 245. How much did he spend?

Step 1: 196 + 245. Step 2: ones 6 + 5 = 11, write 1 and carry 1 ten. Step 3: tens 9 + 4 + 1 = 14, write 4 and carry 1 hundred. Step 4: hundreds 1 + 2 + 1 = 4. Answer: Rs 441.

Iram bought a teddy bear for Rs 384 and a bicycle for Rs 196. How much did she spend?

Step 1: 384 + 196. Step 2: ones 4 + 6 = 10, write 0 and carry 1 ten. Step 3: tens 8 + 9 + 1 = 18, write 8 and carry 1 hundred. Step 4: hundreds 3 + 1 + 1 = 5. Answer: Rs 580.

Unit 2: Number Operations — Subtraction

Subtraction with Borrowing (2-digit)

Opener: 15 pencils minus 8 pencils = 7 fewer pencils. 24 - 8 = 16: borrow 1 ten, ones become 14; 14 - 8 = 6; tens 1 - 0 = 1. Answer 16.

42 - 15 = 27 apples: borrow 1 ten, ones become 12; 12 - 5 = 7; tens 3 - 1 = 2. Answer 27.

Clue words for subtraction: left, how many more, how many less, how many fewer, remaining, difference.

Exercise 1 — Subtraction with Borrowing

23 - 5 = 18; 34 - 8 = 26; 91 - 2 = 89; 40 - 6 = 34; 63 - 27 = 36; 57 - 18 = 39; 95 - 36 = 59; 61 - 42 = 19; 70 - 41 = 29; 82 - 55 = 27; 98 - 59 = 39; 80 - 72 = 8.

A bus had 45 passengers. 18 women got off. How many men are left in the bus?

Step 1: 45 - 18. Step 2: borrow 1 ten, ones become 15; 15 - 8 = 7. Step 3: tens 3 - 1 = 2. Answer: 27 men.

Three-Digit Subtraction without Borrowing

148 - 5 = 143: ones 8 - 5 = 3; tens 4; hundreds 1. 287 - 63 = 224 pages: ones 7 - 3 = 4; tens 8 - 6 = 2; hundreds 2. 475 - 235 = 240 female workers: ones 5 - 5 = 0; tens 7 - 3 = 4; hundreds 4 - 2 = 2.

Key Facts: zero subtracted from a number gives the number itself. A number minus itself is zero.

Exercise 2 — Three-Digit Subtraction without Borrowing

248 - 6 = 242; 309 - 7 = 302; 675 - 4 = 671; 763 - 12 = 751; 845 - 42 = 803; 687 - 231 = 456; 438 - 238 = 200; 786 - 43 = 743; 569 - 307 = 262; 852 - 421 = 431; 701 - 201 = 500; 987 - 876 = 111.

There are 685 students in a school, of whom 384 are girls. How many boys are there?

Step 1: 685 - 384. Step 2: ones 5 - 4 = 1. Step 3: tens 8 - 8 = 0. Step 4: hundreds 6 - 3 = 3. Answer: 301 boys.

Three-Digit Subtraction with Borrowing

132 - 9 = 123 pens: borrow 1 ten, ones become 12; 12 - 9 = 3; tens 2 - 0 = 2; hundreds 1. Answer 123.

Festival: Maryam spent Rs 465 and her friend spent Rs 26 less: 465 - 26 = 439. Ones: borrow 1 ten, ones become 15; 15 - 6 = 9. Tens: 5 - 2 = 3. Hundreds: 4. Answer Rs 439. (Note: the book's Step 1 box on this page

misprints '12 ones - 9 ones = 3 ones', which does not match this problem; the correct ones step is 15 - 6 = 9, and the book's own final answer is 439.)

502 - 245 = 257 apple trees (borrowing across zero): ones 2 - 5 needs a borrow, tens is 0, so borrow 1 hundred; hundreds 5 becomes 4, tens becomes 10, then borrow 1 ten for the ones; ones become 12; 12 - 5 = 7; tens 9 - 4 = 5; hundreds 4 - 2 = 2. Answer 257.

Exercise 3 — Three-Digit Subtraction with Borrowing

143 - 6 = 137; 564 - 7 = 557; 302 - 5 = 297; 651 - 9 = 642; 740 - 3 = 737; 562 - 34 = 528; 284 - 56 = 228; 445 - 57 = 388; 760 - 71 = 689; 832 - 78 = 754; 701 - 54 = 647; 342 - 158 = 184; 452 - 154 = 298; 711 - 267 = 444; 803 - 289 = 514; 900 - 612 = 288.

A train had 658 passengers. 269 passengers got off. How many passengers are left?

Step 1: 658 - 269. Step 2: borrow 1 ten, ones become 18; 18 - 9 = 9. Step 3: tens 4 - 6 needs a borrow, borrow 1 hundred; tens become 14; 14 - 6 = 8. Step 4: hundreds 5 - 2 = 3. Answer: 389 passengers left.

Mixed Addition and Subtraction

In a zoo there are 528 birds and 395 animals. How many more birds are there than animals, and how many are there in all?

More birds: 528 - 395 = 133 (borrow across: ones 8 - 5 = 3; tens 2 - 9 needs a borrow, tens become 12; 12 - 9 = 3; hundreds 4 - 3 = 1). Total: 528 + 395 = 923 (ones 8 + 5 = 13, write 3 and carry 1 ten; tens 2 + 9 + 1 = 12, write 2 and carry 1 hundred; hundreds 5 + 3 + 1 = 9). Answers: 133 more birds; 923 in all.

A bookseller had 385 books. He bought 145 more books and sold 265 books.

How many books are left?

Step 1: 385 + 145 = 530. Step 2: 530 - 265 = 265 (ones 0 - 5 needs a borrow, ones become 10; 10 - 5 = 5; tens 2 - 6 needs a borrow, tens become 12; 12 - 6 = 6; hundreds 4 - 2 = 2). Answer: 265 books left.

Mental Subtraction Strategy

26 - 10 = 16: split 10 into 10 + 0; 26 - 10 = 16. 39 - 12 = 27: split 12 into 10 + 2; 39 - 10 = 29; 29 - 2 = 27.

Practice: 34 - 20 = 14; 28 - 13 = 15; 20 - 10 = 10; 30 - 20 = 10; 50 - 30 = 20; 15 - 10 = 5; 27 - 12 = 15; 36 - 11 = 25; 38 - 17 = 21; 42 - 22 = 20; 49 - 14 = 35. (Note: one practice line on the book's page misprints '34 - 12' under a 34 - 20 split; the split and the answer 14 belong to 34 - 20.)

Review — Subtraction

Multiple-choice answers: first subtract the ones (a). 500 - 300 = 200 (b). 100 - 10 = 90 (a). A number minus itself is zero (a). 18 - 0 = 18 (c).

Subtraction drill: (a) 53 - 8 = 45. (b) 80 - 5 = 75. (c) 72 - 27 = 45. (d) 96 - 19 = 77. (e) 61 - 37 = 24. (f) 51 - 39 = 12. (g) 49 - 20 = 29. (h) 68 - 54 = 14. (i) 619 - 209 = 410. (j) 131 - 5 = 126. (k) 780 - 71 = 709. (l) 500 - 55 = 445. (m) 349 - 149 = 200. (n) 817 - 408 = 409. (o) 511 - 312 = 199. (p) 750 - 169 = 581. (q) 905 - 509 = 396.

Umer had 42 toys and distributed 18 toys. How many toys are left with Umer?

Step 1: 42 - 18. Step 2: borrow 1 ten, ones become 12; 12 - 8 = 4. Step 3: tens 3 - 1 = 2. Answer: 24 toys.

A factory made 624 bicycles and sold 435 bicycles. How many bicycles are remaining?

Step 1: 624 - 435. Step 2: ones 4 - 5 needs a borrow, ones become 14; 14 - 5

= 9. Step 3: tens 1 - 3 needs a borrow, tens become 11; 11 - 3 = 8. Step 4: hundreds 5 - 4 = 1. Answer: 189 bicycles.

Sana got Rs 850 as Eidi and gave Rs 375 to Ahmad. How much money is left with Sana?

Step 1: 850 - 375. Step 2: ones 0 - 5 needs a borrow, ones become 10; 10 - 5 = 5. Step 3: tens 4 - 7 needs a borrow, tens become 14; 14 - 7 = 7. Step 4: hundreds 7 - 3 = 4. Answer: Rs 475.

A train has 965 seats. 780 seats are occupied. How many seats are vacant?

Step 1: 965 - 780. Step 2: ones 5 - 0 = 5. Step 3: tens 6 - 8 needs a borrow, tens become 16; 16 - 8 = 8. Step 4: hundreds 8 - 7 = 1. Answer: 185 vacant seats.

Unit 2: Number Operations — Multiplication

Repeated Addition

Opener: 3 + 3 + 3 = 9 fishes. Repeated addition is multiplication, written with the symbol x (times). 4 x 2 = 8; 6 x 5 = 30; 5 x 3 = 15.

Counting in steps: 2s: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. 3s: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. 4s: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. 5s: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. 10s: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100.

Exercise 1 — Repeated Addition

Stars: 4 rows of 3 stars = 4 x 3 = 12. Flowers: 5 pots with 5 flowers each = 5 x 5 = 25. Cherries: 7 pairs, 2 in each pair = 7 x 2 = 14.

Sweets: (a) 3 x 2 = 6. (b) 3 x 5 = 15. (c) 4 x 4 = 16. (d) 2 x 10 = 20. (Counts verified from the pictures.)

Fill in the blanks: 5 x 2 = 10; 6 x 4 = 24; 5 x 5 = 25; 3 x 10 = 30.

Tables of 2, 3, 4, 5 and 10

Table of 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20. Table of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. Table of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40. Table of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. Table of 10: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100.

3 pots with 4 flowers each = 3 x 4 = 12 flowers. Clue words for multiplication: Product, In all, Times, Altogether. Try Yourself: 6 x 4 = 24.

Exercise 2 — Multiplication Tables

Matching: 2 x 3 = 6; 5 x 3 = 15; 8 x 3 = 24; 10 x 3 = 30; 6 x 3 = 18; 3 x 10 = 30; 5 x 10 = 50; 10 x 10 = 100; 7 x 10 = 70; 8 x 10 = 80; 5 x 2 = 10; 8 x 2 = 16; 2 x 2 = 4; 9 x 2 = 18; 6 x 2 = 12.

Wheels of 4: 4 x 1 = 4; 4 x 2 = 8; 4 x 3 = 12; 4 x 4 = 16; 4 x 5 = 20; 4 x 6 = 24; 4 x 7 = 28 (given); 4 x 8 = 32; 4 x 9 = 36; 4 x 10 = 40. Wheels of 5: 5 x 1 = 5; 5 x 2 = 10; 5 x 3 = 15; 5 x 4 = 20; 5 x 5 = 25 (given); 5 x 6 = 30; 5 x 7 = 35; 5 x 8 = 40; 5 x 9 = 45; 5 x 10 = 50.

More products: (a) 3 x 3 = 9 (given). (b) 5 x 2 = 10. (c) 4 x 5 = 20. (d) 1 x 10 = 10. (e) 7 x 4 = 28. (f) 7 x 3 = 21. (g) 10 x 4 = 40. (h) 10 x 10 = 100. (i) 8 x 2 = 16.

Column multiplication: (a) 3 x 5 = 15 (given). (b) 6 x 4 = 24. (c) 4 x 2 = 8. (d) 6 x 3 = 18. (e) 9 x 4 = 36. (f) 8 x 5 = 40. (g) 9 x 3 = 27. (h) 10 x 2 = 20. (i) 4 x 4 = 16. (j) 7 x 2 = 14. (k) 10 x 5 = 50. (l) 10 x 10 = 100.

A cat has 4 kittens. How many kittens do 6 cats have?

Step 1: 6 cats x 4 kittens = 6 x 4. Step 2: 6 x 4 = 24. Answer: 24 kittens.

A flower has 10 petals. How many petals do 9 flowers have?

Step 1: 10 x 9 = 90. Answer: 90 petals.

A bicycle has 2 wheels. How many wheels do 4 bicycles have?

Step 1: 4 x 2 = 8. Answer: 8 wheels.

A spider has 8 legs. How many legs do 5 spiders have?

Step 1: 5 x 8 = 40. Answer: 40 legs.

Review — Multiplication

Multiple-choice answers: 2 + 2 + 2 + 2 = 4 times 2 (c). 3, 6, 9, 12, 15 is the table of 3 (c). 10 x 5 = 50 (d). 7 times 3 = 21 (d). 4, 8, 12, 16, 20, 24 continues with 20 at the missing place (d).

Balloons: 6 x 3 = 18; 4 x 4 = 16; 3 x 5 = 15; 7 x 2 = 14. Steps of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30. Steps of 10: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100.

Matching: 9 x 3 = 27; 5 x 9 = 45; 5 x 10 = 50; 7 x 5 = 35; 6 x 2 = 12; 10 x 2 = 20; 7 x 3 = 21; 8 x 4 = 32; 9 x 2 = 18; 10 x 10 = 100.

Products: (a) 7 x 2 = 14. (b) 4 x 3 = 12. (c) 5 x 5 = 25. (d) 6 x 4 = 24. (e) 8 x 10 = 80. (f) 9 x 4 = 36. (g) 1 x 2 = 2. (h) 8 x 3 = 24. (i) 6 x 3 = 18. (j) 1 x 10 = 10. (k) 8 x 5 = 40. (l) 9 x 10 = 90.

A packet has 10 pencils. How many pencils are there in 4 packets?

Step 1: 4 x 10 = 40. Answer: 40 pencils.

There are 8 flowers in a flowerpot. How many flowers are there in 3 flowerpots?

Step 1: 3 x 8 = 24. Answer: 24 flowers.

There are 5 oranges in a basket. How many oranges are there in 7 baskets?

Step 1: 7 x 5 = 35. Answer: 35 oranges.

A bird has 2 legs. How many legs do 7 birds have?

Step 1: 7 x 2 = 14. Answer: 14 legs.

Unit 2: Number Operations — Division

Division as Equal Sharing

Opener: 12 biscuits shared in 3 boxes = 12 / 3 = 4 biscuits in each box.

Successive subtraction: 6 balloons among 3 children: 6 - 3 - 3 = 0, so each child gets 2 balloons. 10 apples between 2 children: 10 - 2 - 2 - 2 - 2 - 2 = 0, so each child gets 5 apples.

Try Yourself: 20 eggs among 4 children = 5 eggs each. 12 balloons among 3 children = 4 balloons each.

8 carrots among 4 rabbits: 8 - 4 - 4 = 0, so 8 / 4 = 2 carrots each. Recall: 2 x 4 = 8.

Key Facts: division is successive subtraction; the symbol for division is /.

Exercise 1 — Division

15 flowers in 3 pots = 15 / 3 = 5 flowers per pot. 24 pencils in 4 boxes = 24 / 4 = 6 pencils per box. 20 ice-creams among 10 children = 20 / 10 = 2 ice-creams per child.

Groups: 8 / 4 = 2 (given); 15 / 3 = 5; 24 / 6 = 4; 18 / 3 = 6.

Divisions: 12 / 2 = 6; 28 / 4 = 7; 40 / 5 = 8; 20 / 2 = 10; 70 / 10 = 7; 45 / 5 = 9; 100 / 10 = 10; 30 / 5 = 6; 27 / 3 = 9; 32 / 8 = 4; 50 / 5 = 10; 4 / 4 = 1.

Key Fact: a number divided by itself is 1.

Sara shared 21 cupcakes equally among 7 friends. How many cupcakes did each friend get?

Step 1: 21 / 7 = 3. Answer: 3 cupcakes each.

32 oranges are packed equally in 4 baskets. How many oranges are in each basket?

Step 1: 32 / 4 = 8. Answer: 8 oranges per basket.

Clue Words for Number Stories

Addition clues: Total, Altogether, In all, Sum, Added to. Subtraction clues: Left,

More than, How many less, How many fewer, Remain, Difference.

Multiplication clues: Product, Times, In all, Altogether. Division clues: How many will each get, How many in each group, Shared, Divided, Equal, Equally.

Steps to solve a number story: read carefully; underline the clue words; draw a picture if needed; write a number sentence; solve it.

A tailor stitched 65 suits in one week and 58 suits in the next week. How many suits did he stitch altogether?

Clue 'altogether' means addition. Step 1: 65 + 58. Step 2: ones 5 + 8 = 13, write 3 and carry 1 ten. Step 3: tens 6 + 5 + 1 = 12. Answer: 123 suits.

Ahmed had Rs 500. He spent Rs 225. How much money is left with him?

Clue 'left' means subtraction. Step 1: 500 - 225. Step 2: ones 0 - 5 needs a borrow, ones become 10; 10 - 5 = 5. Step 3: tens 9 - 2 = 7 (after borrowing across zero). Step 4: hundreds 4 - 2 = 2. Answer: Rs 275 left.

Ahmed has 5 books in a bag. How many books are there in 6 bags?

Reason: equal groups, so we multiply. Step 1: 5 x 6 = 30. Answer: 30 books.

27 bananas are shared equally among 3 monkeys. How many bananas does each monkey get?

Clue 'equally' means division. Step 1: 27 / 3 = 9. Answer: 9 bananas each.

Review — Division

I have learnt to: divide by successive subtraction; share equally; use the division symbol; recall that a number divided by itself is 1 and a number divided by 1 is the number itself.

Vocabulary: Divide, Successive subtraction, Equally sharing.

Multiple-choice answers: division is successive subtraction (c). The symbol / is for division (d). 100 / 10 = 10 (d). A number divided by 1 is the number itself

(d). 5 / 5 = 1 (b).

Successive subtraction: 12 - 4 = 8; 8 - 4 = 4; 4 - 4 = 0, so 12 / 4 = 3. 20 - 5 = 15; 15 - 5 = 10; 10 - 5 = 5; 5 - 5 = 0, so 20 / 5 = 4.

Sharing: 10 / 5 = 2 pigeons; 6 / 3 = 2 toys; 8 / 2 = 4 balls; 12 / 4 = 3 rings.

Train divisions: 10 / 10 = 1; 8 / 4 = 2; 6 / 2 = 3; 20 / 5 = 4; 40 / 10 = 4; 25 / 5 = 5; 28 / 4 = 7; 24 / 3 = 8; 36 / 4 = 9.

Ali shared 30 chocolates equally among 5 friends. How many chocolates did each friend get?

Step 1: 30 / 5 = 6. Answer: 6 chocolates each.

Ramsha shared 20 suits equally among 2 children. How many suits did each child get?

Step 1: 20 / 2 = 10. Answer: 10 suits each child gets.

Unit 3: Fractions

Equal Parts

Opener: Amna divides her birthday cake into three equal parts and gives them to her friends.

Hina divides a cake into two equal parts; Raza divides a cake into two unequal parts. Who divided the cake in equal parts? Hina.

Key Fact: Equal parts have the same shape and the same size.

Shapes in unequal parts: triangle with uneven horizontal strips, rectangle with uneven divisions, oval with uneven divisions, circle with uneven divisions, trapezium with uneven divisions. Shapes in equal parts: square divided into 4 equal squares, circle divided into 2 equal halves, hexagon divided into equal parts, triangle divided into 2 equal parts, rectangle divided into 3 equal strips.

One-Half

A whole pizza divided into two equal parts; each equal part is called one-half.

Key Fact: Two halves together make one whole.

Colour one half of: the circle (2 equal halves), the triangle (2 equal halves), the oval (2 equal halves), the hexagon (2 equal halves), the heart (2 equal halves). Do NOT colour: the rectangle divided into 3 strips (thirds, not halves), the quadrilateral divided by a diagonal into unequal parts, the rectangle divided by diagonals into 4 triangles.

Try Yourself — Equal Parts

Tick the objects divided into equal parts: biscuit split into 2 equal halves — tick. Banana cut diagonally into 2 unequal parts — no tick. Two apple halves, equal — tick. Chocolate bar, one big piece and one small piece, unequal — no tick.

Tick the shapes divided into equal parts (verified at 200 DPI): hexagon (4 equal parts) — tick; square (4 equal squares) — tick; oval (2 equal halves) — tick; rectangle (3 equal strips) — tick; heart (2 equal halves) — tick. No tick: circle (divided horizontally above the centre, so the two parts are unequal), triangle (3 parts from an interior point below the centre, unequal), trapezium (2 unequal parts).

One-Quarter and One-Third

A whole pizza divided into two equal parts, then further divided into four equal parts; each equal part is called one-quarter. Key Fact: Four quarters together make one whole.

Colour one quarter of: the circle (4 equal quarters), the hexagon (4 equal parts), the square (4 equal squares), the rectangle divided into 4 equal strips.

Do NOT colour: the rectangle divided into 3 strips, the heart divided into 2 halves.

A whole cake divided into three equal parts; each equal part is called one-third. Key Fact: Three thirds together make one whole.

Colour one third of: the rectangle divided into 3 equal strips, the triangle divided into 3 equal parts, the circle divided into 3 equal sectors, the oval divided into 3 equal strips, the rectangle divided into 3 equal horizontal strips. Do NOT colour: the square divided into 4 parts.

Fraction in Numerical Form

A triangle has two equal parts; one part coloured = one-half, written as 1/2. A circle has three equal parts; one part coloured = one-third, written as 1/3. A square has four equal parts; one part coloured = one-quarter, written as 1/4.

Colour one part and write the fraction: circle (2 parts) = 1/2 (given); triangle (3 parts) = 1/3; square (4 parts) = 1/4; rectangle (3 strips) = 1/3; circle (4 parts) = 1/4; heart (2 parts) = 1/2.

Fractions from 1/2 to 1/10: 2 parts = One-half = 1/2; 3 parts = One-third = 1/3; 4 parts = One-fourth = 1/4; 5 parts = One-fifth = 1/5; 6 parts = One-sixth = 1/6; 7 parts = One-seventh = 1/7; 8 parts = One-eighth = 1/8; 9 parts = One-ninth = 1/9; 10 parts = One-tenth = 1/10.

Can two quarters together make a whole? No. Four quarters together make one whole; two quarters together make one-half.

More About Fractions

Two-thirds = 2/3. Three-fourths = 3/4. Four-fifths = 4/5. Five-sixths = 5/6. Six-sevenths = 6/7. Seven-eighths = 7/8. Eight-ninths = 8/9. Nine-tenths = 9/10.

Fraction of coloured parts = Number of coloured parts / Total number of equal parts.

Exercise 1 — Fractions

Match the shape with the fraction: (a) 1/3 -> triangle (3 parts, 1 coloured); 1/4 -> circle (4 quarters, 1 coloured); 1/2 -> oval (2 halves, 1 coloured); 1/5 -> rectangle (5 strips, 1 coloured).

Match (b): 1/7 -> horizontal rectangle (7 strips, 1 coloured); 1/9 -> triangle row (9 small triangles, 1 coloured); 1/8 -> circle (8 sectors, 1 coloured). Note: the fourth shape in (b), a vertical strip rectangle, is drawn with about 10 narrow strips, so it corresponds to 1/10, not to the printed 1/6; the printed fraction 1/6 has no matching shape in the set, and the strip rectangle has no matching fraction in the printed list. Counts verified at 200 DPI.

Colour one part and write the fraction (counts verified at 200 DPI): (a) rectangle, 3 strips = 1/3 (given). (b) hexagon, 4 parts = 1/4. (c) triangle, 2 parts = 1/2. (d) flower, 5 petals = 1/5. (e) circle, 6 sectors = 1/6. (f) zigzag strip, 5 segments = 1/5. (g) big triangle, 9 small triangles = 1/9. (h) square, 8 triangular parts = 1/8. (i) chevron strip, 10 parts = 1/10.

Exercise 2 — Fractions

Write the fraction for the coloured part (counts verified at 200 DPI): (a) rectangle, 5 strips, 2 coloured = 2/5 (given). (b) rectangle, 4 strips, 3 coloured = 3/4. (c) rectangle, 4 strips, 3 coloured = 3/4. (d) circle, 6 sectors, 2 coloured = 2/6. (e) circle, 8 sectors, 4 coloured = 4/8. (f) circle, 7 sectors, 3 coloured = 3/7. (g) staircase shape, 9 squares (4 top + 5 bottom), 4 coloured = 4/9. (h) rectangle, 10 squares, 3 coloured = 3/10. (i) rectangle, 10 squares, 7 coloured = 7/10.

Colour the shape for the given fraction: (a) pentagon, 5 parts, colour 3 = 3/5. (b) hexagon, 6 parts, colour 5 = 5/6. (c) rectangle, 4 triangles, colour 3 = 3/4. (d) grid, 8 squares, colour 7 = 7/8. (e) shape, 7 squares, colour 2 = 2/7. (f) grid, 10 squares, colour 5 = 5/10. (g) circle, 10 sectors, colour 6 = 6/10. (h) circle, 9 sectors, colour 7 = 7/9. (i) circle, 10 sectors, colour 9 = 9/10.

Review — Fractions

I have learnt to: recognise fractions as equal parts of a whole; identify half, one third and quarter with objects and figures; represent half, one third and quarter in numerical form (1/2, 1/3 and 1/4); shade the equal parts of a given figure to match a given fraction; recognise and name unit fractions up to 1/10; recognise fractions like two thirds (2/3), three fourths (3/4), four fifths (4/5), up to nine tenths (9/10).

Vocabulary: Fraction, One-half, One-third, One-quarter.

Multiple-choice answers: (i) One-quarter is written as 1/4 (c). (ii) Shape with 7 strips, 4 coloured = 4/7 (d). (iii) Equal parts have the same shape and the same size (b). (iv) 4/5 is called four-fifths (a). (v) Fraction of coloured parts = Number of coloured parts / Total number of equal parts (d).

Colour the shapes divided into halves: star (2 equal halves) — colour; oval (2 equal halves) — colour; triangle (2 equal halves) — colour. Do NOT colour: square (4 quarters), circle (divided off-centre into unequal parts, verified at 200 DPI).

Match the coloured shapes with the fraction (counts verified at 200 DPI): pink circle, 9 sectors, 8 coloured = 8/9. Blue circle, 4 quarters, 3 coloured = 3/4. Yellow circle, 5 sectors, 1 coloured = 1/5. Yellow circle, 6 sectors, 5 coloured = 5/6. Purple circle, 10 sectors, 7 coloured = 7/10. Green circle, 7 sectors, 2 coloured = 2/7.

Colour for the fraction: (a) hexagon, 2 parts, colour 1 = 1/2. (b) triangle, 3 parts, colour 1 = 1/3. (c) diamond, 4 parts, colour 1 = 1/4. (d) circle, 6 sectors, colour 5 = 5/6. (e) star, 5 parts, colour 3 = 3/5. (f) triangle, 9 small triangles, colour 7 = 7/9.

Write the fraction for the coloured part (counts verified at 200 DPI): (a) strip, 5 squares, 3 coloured = 3/5. (b) strip, 7 squares, 4 coloured = 4/7. (c) diamond, 4 parts, 2 coloured = 2/4. (d) zigzag, 7 segments, 4 coloured = 4/7. (e) circle, 8 sectors, 4 coloured = 4/8. (f) grid, 10 squares, 5 coloured = 5/10.

Unit 4: Measurement — Length

Metre and Centimetre

Opener: 'My mother is taller than I. Is there any instrument with which we can measure our heights?' Yes — a measuring tape or a metre rod.

The car is longer than the motorbike. To measure the exact length of objects, we need the standard units of length.

Metre: Metre is the standard unit of length. The symbol 'm' is used for metre. The length of rope, wire, ribbon, table, etc., are measured in metres. Key Fact: The length, width and height of longer objects are measured in metres.

Centimetre: The symbol 'cm' is used for centimetre. The ruler is divided into 15 equal parts; the length of each part is 1 cm. The length of pencil, notebook, chocolate, etc., are measured in centimetres. The length of chocolate is 8 cm. Key Facts: Centimetres are used to measure the lengths of shorter objects. 1 m = 100 cm.

Class Activity: Measure and record the lengths of classroom objects using a metre rod and a ruler (longest and shortest object — student activity, no fixed key).

Exercise 1 — Length

Metres (m) or centimetres (cm)? (i) The length of a table = m. (ii) The length of your notebook = cm. (iii) The length of school bus = m. (iv) The length of cricket bat = m (classified as a longer object under the book's rule; a bat is about 85 cm long, near a metre — a judgement call, not book-stated). (v) The length of your lunch box = cm. (vi) The length of a tissue box = cm.

Lengths from the ruler (guide lines read at 200 DPI): (i) The length of the pencil = 9 cm. (ii) The length of the key = 6 cm. (iii) The length of the toothbrush = 12 cm. (iv) The length of the eraser = 3 cm.

Comparisons (any true comparison is correct; natural completions): (i) The pencil is longer than the key. (ii) The toothbrush is longer than the pencil. (iii) The key is shorter than the toothbrush. (iv) The eraser is shorter than the key.

Try Yourself: In your school bag, find an object smaller than your eraser and measure its length (student activity).

Measure the printed pictures with a ruler (lengths as printed in the PDF at 100% scale, so they depend on print size): sharpener = about 2 cm; pen = about 6 cm; crayon = about 3 cm. Note: this is a hands-on measuring activity — the reading depends on the print size of the book, so these values are approximate, not a fixed key.

Addition and Subtraction of Lengths

Red ribbon 25 cm + green ribbon 18 cm = 43 cm total length of both ribbons.

Red ribbon longer than green: 25 cm - 18 cm = 7 cm longer.

Observe: 53 m + 29 m = 82 m. 72 cm - 34 cm = 38 cm.

Exercise 2 — Lengths

Add: (a) 10 m + 8 m = 18 m. (b) 16 m + 14 m = 30 m. (c) 39 cm + 24 cm = 63 cm. (d) 50 cm + 17 cm = 67 cm. (e) 48 m + 43 m = 91 m. (f) 67 cm + 27 cm = 94 cm. (g) 79 m + 12 m = 91 m. (h) 89 cm + 8 cm = 97 cm.

Subtract: (a) 15 m - 6 m = 9 m. (b) 28 m - 12 m = 16 m. (c) 57 cm - 28 cm = 29 cm. (d) 40 cm - 13 cm = 27 cm. (e) 71 m - 32 m = 39 m. (f) 69 cm - 19 cm = 50 cm. (g) 85 m - 58 m = 27 m. (h) 98 cm - 29 cm = 69 cm.

Nadia bought 18 m of white cloth and 15 m of green cloth. How much cloth did Nadia buy altogether?

Step 1: 18 + 15. Step 2: ones 8 + 5 = 13, write 3 and carry 1 ten. Step 3: tens 1 + 1 + 1 = 3. Answer: Nadia bought 33 m cloth altogether.

The length of Ahmad's lunch box is 24 cm and the length of his brother's lunch box is 18 cm. Which lunch box is longer and how much?

Step 1: 24 - 18. Step 2: borrow 1 ten, ones become 14; 14 - 8 = 6. Step 3: tens 1 - 1 = 0. Answer: Ahmad's lunch box is longer by 6 cm.

Unit 4: Measurement — Mass

Kilogram and Gram

To measure the accurate mass of objects, we need the standard unit of mass.

Kilogram: The standard unit of mass is kilogram. The symbol 'kg' is used for kilogram. It is used to measure the mass of heavy objects like flour bag, watermelon, etc. The mass of watermelon is 6 kg; the mass of flour is 10 kg. So, the mass of flour bag is 4 kg more than the watermelon (10 - 6 = 4). Key Fact: Kilogram is used to measure the mass of heavy objects.

Gram: Gram is the unit of mass. The symbol 'g' is used for gram. It is used to measure the mass of lighter objects like pencil, biscuit, etc. The mass of biscuit pack is 300 g. Key Facts: Gram is used to measure the mass of lighter objects. 1 kg = 1000 g.

Class Activity: Measure and record the mass of school bags (heaviest and lightest — student activity).

Exercise 3 — Mass

Kilograms (kg) or grams (g)? (i) The mass of a biscuit = g. (ii) The mass of apples (pile pictured) = kg. (iii) The mass of a geometry (box) = g. (iv) The mass of Mangos (basket pictured) = kg. (v) The mass of a sugar bag = kg.

Read the scales: milk/juice carton = 250 g; pumpkin = 2 kg; mango = 80 g; tomatoes = 4 kg.

Try Yourself: Salt = 500 g, watermelon = 2 kg. Watermelon is heavier (2 kg = 2000 g, which is more than 500 g).

Addition and Subtraction of Masses

Father bought 20 kg apples + 15 kg guavas = 35 kg total mass of fruits.

Apples more than guavas: 20 kg - 15 kg = 5 kg more.

Observe: 45 g + 17 g = 62 g. 66 g - 28 g = 38 g.

Exercise 4 — Masses

Add: (a) 16 kg + 9 kg = 25 kg. (b) 27 kg + 19 kg = 46 kg. (c) 40 g + 30 g = 70 g. (d) 53 g + 27 g = 80 g. (e) 63 kg + 29 kg = 92 kg. (f) 46 g + 36 g = 82 g. (g) 72 g + 27 g = 99 g. (h) 85 kg + 9 kg = 94 kg.

Subtract: (a) 18 kg - 9 kg = 9 kg. (b) 35 kg - 16 kg = 19 kg. (c) 42 g - 23 g = 19 g. (d) 60 g - 25 g = 35 g. (e) 67 kg - 29 kg = 38 kg. (f) 51 g - 22 g = 29 g. (g) 88 g - 49 g = 39 g. (h) 99 kg - 36 kg = 63 kg.

Maryam bought 60 g red pepper and 35 g black pepper. Tell the total mass of both peppers.

Step 1: 60 + 35. Step 2: ones 0 + 5 = 5. Step 3: tens 6 + 3 = 9. Answer: The

total mass of both peppers is 95 g.

A grocer buys 85 kg potatoes. He sells 48 kg potatoes. How much potatoes are left?

Step 1: 85 - 48. Step 2: borrow 1 ten, ones become 15; 15 - 8 = 7. Step 3: tens 7 - 4 = 3. Answer: 37 kg potatoes are left.

Unit 4: Measurement — Capacity

Capacity and Its Standard Units

One jug fills three glasses of water. The jug holds more water; the glass holds less water than the jug. The bucket holds more water than the bowl. Key Fact: The larger the container, the more capacity it will have.

Try Yourself: Tick the container with more capacity: bottle (vs glass) — tick; kettle (vs cup) — tick; big juice carton (vs small milk carton) — tick; tub (vs small pot) — tick.

Litre: Litre is the standard unit of capacity. The symbol 'l' (litre) is used for litre. Water, milk and petrol are measured in litres.

Millilitre: Millilitre is also the unit of capacity. The symbol 'ml' is used for millilitre. It is used to measure the capacity of small containers. The capacity of glass, cup, etc., are measured in millilitres.

Key Facts: The capacity of large containers is measured in litres and the capacity of small containers is measured in millilitres. 1 litre = 1000 millilitres.

Exercise 5 — Capacity

Litres or millilitres? (i) The capacity of a tub = litres (given). (ii) The capacity of a cup = millilitres. (iii) The capacity of a spoon = millilitres. (iv) The capacity of a jug = litres. (v) The capacity of an inkpot = millilitres.

Encircle the containers that have capacity in litres: oil can, measuring beaker (marked 2 litres), tub. Do NOT encircle: juice carton (millilitres), glass (millilitres).

Addition and Subtraction of Capacity

Cow gives 12 litres milk in the morning + 9 litres in the evening = 21 litres milk in one day.

Morning more than evening: 12 litres - 9 litres = 3 litres more milk in the morning.

Observe: 35 ml + 46 ml = 81 ml. 43 ml - 16 ml = 27 ml.

Exercise 6 — Capacities

Add: (a) 22 ml + 14 ml = 36 ml. (b) 38 litres + 15 litres = 53 litres. (c) 40 ml + 18 ml = 58 ml. (d) 54 ml + 37 ml = 91 ml. (e) 71 litres + 19 litres = 90 litres. (f) 65 ml + 25 ml = 90 ml. (g) 85 litres + 10 litres = 95 litres. (h) 49 ml + 49 ml = 98 ml.

Subtract: (a) 11 ml - 9 ml = 2 ml. (b) 20 litres - 12 litres = 8 litres. (c) 32 ml - 13 ml = 19 ml. (d) 55 ml - 26 ml = 29 ml. (e) 73 litres - 48 litres = 25 litres. (f) 67 ml - 28 ml = 39 ml. (g) 81 litres - 59 litres = 22 litres. (h) 90 ml - 45 ml = 45 ml.

Irfan put 18 litres of petrol on Monday and 14 litres of petrol on Tuesday in his car. How much total petrol did he put in his car in two days?

Step 1: 18 + 14. Step 2: ones 8 + 4 = 12, write 2 and carry 1 ten. Step 3: tens 1 + 1 + 1 = 3. Answer: 32 litres of petrol.

A milkman bought 75 litres of milk and sold 68 litres of milk. How much milk is left with him?

Step 1: 75 - 68. Step 2: borrow 1 ten, ones become 15; 15 - 8 = 7. Step 3:

tens 6 - 6 = 0. Answer: 7 litres of milk left.

Review — Measurement

I have learnt to: compare the length of different objects; recognise the metre and centimetre as the standard units of length; add and subtract the standard units of length; use the standard units of length in real life; compare the mass of different objects; recognise the kilogram and gram as the standard units of mass; add and subtract the standard units of mass; use the standard units of mass in real life; compare the capacity of different objects; recognise the litre and millilitre as the standard units of capacity; add and subtract the standard units of capacity; use the standard units of capacity in real life.

Vocabulary: Length, Metre, Centimetre, Mass, Kilogram, Gram, Capacity, Litre, Millilitre.

Multiple-choice answers: (i) The height of a tree is measured in metres (d). (ii) The standard unit of capacity is litre (b). (iii) The symbol of kilogram is kg (c). (iv) The length of short objects is measured in centimetres (b). (v) Kilogram is the standard unit of mass (d).

Tick the suitable unit: packet of chips -> g; geometry box length -> cm; water tank capacity -> litres; pole height -> m; chair mass -> kg.

Add: (a) 37 cm + 18 cm = 55 cm. (b) 83 kg + 16 kg = 99 kg. (c) 59 ml + 29 ml = 88 ml. (d) 67 g + 13 g = 80 g.

Subtract: (a) 65 ml - 56 ml = 9 ml. (b) 93 litres - 47 litres = 46 litres. (c) 77 kg - 58 kg = 19 kg. (d) 91 m - 22 m = 69 m.

A shopkeeper bought 80 litres of ice-cream. He sold 52 litres of ice-cream in a day. How much ice-cream was left with him?

Step 1: 80 - 52. Step 2: ones 0 - 2 needs a borrow, ones become 10; 10 - 2 =

8. Step 3: tens 7 - 5 = 2. Answer: 28 litres of ice-cream left.

The mass of a wheat bag is 15 kg more than a rice bag. If the mass of the rice bag is 48 kg, then find the mass of the wheat bag.

Step 1: 48 + 15. Step 2: ones 8 + 5 = 13, write 3 and carry 1 ten. Step 3: tens 4 + 1 + 1 = 6. Answer: The mass of the wheat bag is 63 kg.

The length of a red wire is 90 m while the length of a yellow wire is 72 m. Which wire is longer and how much?

Step 1: 90 - 72. Step 2: ones 0 - 2 needs a borrow, ones become 10; 10 - 2 = 8. Step 3: tens 8 - 7 = 1. Answer: The red wire is longer by 18 m.

Unit 5: Time — Hours, Minutes, a.m. and p.m.

Hours and Minutes

Opener: Amna goes to school at 7 o'clock in the morning and does her homework at 5 o'clock in the evening. Morning time is written with a.m. and evening time with p.m.

Ahmad's breakfast clock: hour hand at 8, minute hand at 3 (15 minutes) — the time is 8:15.

The dial of the clock is divided into 12 big parts. Each big part is further divided into 5 equal small parts. One small part represents one minute. The dial is divided into 60 equal small parts.

The minute hand moves from one number to the other number in 5 minutes. When the minute hand completes one round in 60 minutes, the hour hand moves to the next number. So, there are 60 minutes in an hour.

Key Facts: 60 minutes = 1 hour. 24 hours = 1 day. There are 24 hours in a day because the hour hand completes two rounds in a day.

Reading and Writing the Time

Hour hand at 6, minute hand at 2 (10 minutes passed) — the time is 6:10, read as 'six ten'.

1:45 — 'One forty-five'. 3:20 — 'Three twenty'.

Key Fact: When the minute hand is at 12, we write '00' (zero) minutes.

Try Yourself — write the time for each clock (small schematic clocks, readings approximate): 10:05, 9:15, 6:00, 6:05, 6:55, 11:40.

Usage of a.m. and p.m.

24 hours in a day; the hour hand completes two rounds in 24 hours (12 hours from noon to night, 12 hours from night to noon).

a.m. (ante-meridiem) is written with time between 12:00 mid-night and 12:00 noon. p.m. (post-meridiem) is written with time between 12:00 noon and 12:00 mid-night.

Ali goes to school in the morning at 7 o'clock = 7:00 a.m. Ali completes homework in the evening at 7 o'clock = 7:00 p.m.

Fatima's Day — Time in a.m. and p.m.

Fatima wakes up: 6:15 a.m. (hour hand at 6, minute hand at 3). Fatima takes breakfast: 7:30 a.m. (hour hand between 7 and 8, minute hand at 6). Fatima comes back from school: 1:00 p.m. (hour hand at 1, minute hand at 12). Fatima does homework: 4:05 p.m. (hour hand at 4, minute hand at 1). Fatima offers Asr namaz: 4:20 p.m. (hour hand between 4 and 5, minute hand at 4). Fatima plays in the evening: 6:45 p.m. (hour hand past 6, minute hand at 9).

Drawing Hands of the Clock

The small hand shows 'hours' and the big hand shows 'minutes'.

Examples: 4:35 (small hand between 4 and 5, big hand at 7); 1:15 (small

hand past 1, big hand at 3).

Try Yourself — draw hands for: 9:40 (small hand between 9 and 10, big hand at 8); 8:05 (small hand just past 8, big hand at 1); 2:20 (small hand between 2 and 3, big hand at 4); 7:55 (small hand between 7 and 8, big hand at 11); 10:10 (small hand just past 10, big hand at 2); 6:30 (small hand between 6 and 7, big hand at 6); 3:30 (small hand between 3 and 4, big hand at 6); 11:25 (small hand between 11 and 12, big hand at 5).

Key Fact: We do not write a.m. and p.m. with 12 o'clock. We write it as 12:00 noon or 12:00 mid-night.

Unit 5: Time — Calendars

Solar Calendar

There are 365 or 366 days in a solar year. There are 12 months in the year. A calendar is the record of all months, dates and days of the year.

The printed calendar's day alignments match the year 2021 (January 1 = Friday); the book does not print the year on the page.

Try Yourself — days for the given dates using the calendar: 10 January = Sunday (given). 23 March = Tuesday. 5 October = Tuesday. 6 September = Monday. 22 June = Tuesday. 5 February = Friday. 14 August = Saturday. 31 July = Saturday. 9 November = Tuesday. 1 May = Saturday. 25 December = Saturday. 21 April = Wednesday.

Also: encircle the date and month of your birthday on the calendar; tick the date and month of the independence of Pakistan (14 August).

Key Facts: The Earth completes one revolution around the sun in 365 days, so there are 365 days in a year. February is the shortest month of the year (28 or 29 days). There are 30 or 31 days in a solar month in general.

Lunar Calendar

There are 354 or 365 days in a lunar year (book's wording — note: a lunar year is 354 or 355 days; '365' here appears to be a misprint for 355). There are 12 months in the year.

Lunar months in order: Muharram, Safar, Rabi-ul-Awwal, Rabi-ul-Sani, Jammadi-ul-Awwal, Jammadi-ul-Sani, Rajab, Sha'ban, Ramadan, Shawwal, Zul-Qadah, Zul-Hajjah.

Try Yourself — days for the given dates using the calendar: 10 Muharram = Monday (given). 18 Jammadi-ul-Awwal = Monday. 5 Safar = Thursday. 20 Jammadi-ul-Sani = Friday. 29 Rajab = Monday. 25 Rabi-ul-Sani = Sunday. 27 Ramadan = Tuesday. 12 Rabi-ul-Awwal = Saturday. 1 Shawal = Friday. 19 Zul-Qadah = Thursday. 10 Zul-Hajjah = Wednesday. 25 Sha'ban = Saturday.

Also: encircle the months in which we celebrate Eids (Shawwal — Eid-ul-Fitr; Zul-Hajjah — Eid-ul-Azha); tick the month in which Muslims fast (Ramadan).

Key Facts: The lunar calendar is also known as the Hijri and Islamic calendar. There are 29 or 30 days in a lunar month, depending upon the sighting of the new moon.

Review — Time

I have learnt to: recognise hours in a day and minutes in an hour; read and write time from a clock in hours and minutes (five-minute intervals); recognise a.m. and p.m.; draw clock hands to show time in hours and minutes (five-minute intervals); use Solar calendar to find a particular date/day; use Islamic calendar to find a particular date/day.

Vocabulary: p.m. and a.m., Solar year, Lunar year, Calendar.

Multiple-choice answers: (i) 1 hour = 60 minutes (d). (ii) The hour hand completes 2 rounds in a day (b). (iii) We write a.m. with time between 12:00

mid-night and 12:00 noon (a). (iv) In the solar year, the shortest month is February (c). (v) Muslims fast in the month of Ramadan (d).

Clock readings (verified at 200 DPI): (a) 10:35. (b) 2:40. (c) 9:45. (d) 6:55. (e) 1:20. (f) 11:25.

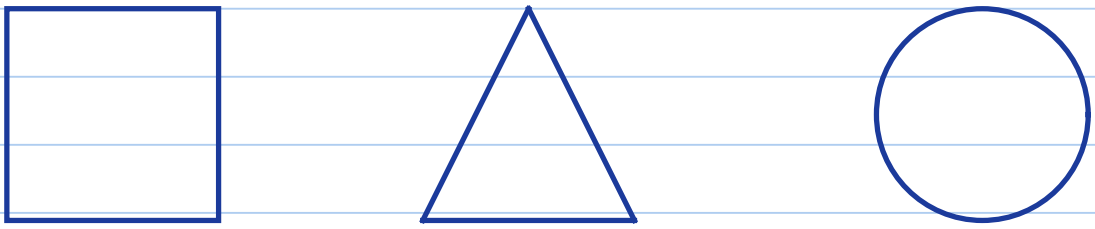
Draw hands for: (a) 3:40 (small hand between 3 and 4, big hand at 8). (b) 8:50 (small hand between 8 and 9, big hand at 10). (c) 10:10 (small hand just past 10, big hand at 2). (d) 2:20 (small hand between 2 and 3, big hand at 4). (e) 7:55 (small hand between 7 and 8, big hand at 11). (f) 12:00 (both hands at 12).

a.m./p.m. by sentence: (a) Ahmad goes to school at 8 o'clock = 8:00 a.m. (b) Our family takes dinner at 9:15 = 9:15 p.m. (c) Ayesha wakes up in the morning at 5:30 = 5:30 a.m. (d) Badar plays football in the evening at 4:45 = 4:45 p.m. (e) Maryam offers Maghrib prayer at 6 o'clock = 6:00 p.m. (f) Hina takes breakfast at 7:25 = 7:25 a.m.

Month order: Solar — January 1st, April 4th, May 5th (given), August 8th, October 10th, November 11th. Lunar — Safar 2nd, Rabi-ul-Awwal 3rd (given), Jammadi-ul-Sani 6th, Rajab 7th, Ramadan 9th, Zul-Hajjah 12th.

Unit 6: Geometry — Shapes

Shapes in the Picture



Square Triangle Circle

Complete the table by writing the name and number of shapes in the picture: Square: 4 (2 house windows + 2 in the bottom row). Circle: 5 (round tree top + sun + 2 car wheels + 1 in the bottom row). Rectangle: 10 (4 tree trunks + car body + car cabin middle + house body + house door + 2 in the bottom row). Triangle: 13 (left tree 2 + middle tree 1 + right tree 1 + car cabin sides 2 + house roof 1 + bottom row 6).

Note: The lower part of the left tree is drawn with a flat top (a truncated triangle / trapezoid). It is counted here as a triangle to fit the book's four-shape table; counted strictly, triangles = 12.

Semi-Circle and Quarter-Circle

Divide a circle into two equal parts; one half part of the circle is called a semi-circle.

Divide the circle into four equal parts; one quarter part of the circle is called a quarter-circle.

Try Yourself — Circles, Semi-Circles, Quarter-Circles

Colour the circle red, the semi-circle green and the quarter-circle yellow.

Totals: Circles: 8 (locomotive wheels 3 + middle figure 1 + crab eyes 2 + crab feet 2). Semi-circles: 4 (glass bowl + glass base + locomotive dome + crab body). Quarter-circles: 9 (locomotive front 1 + middle figure 4 + bottom dome halves 2 + crab claws 2).

Note: The locomotive's blue body also has two white semi-circular cutouts; if those are counted as semi-circles, semi-circles = 6 instead of 4.

Sides and Vertices of the Shapes

Rectangle: 4 sides and 4 vertices. Key Facts: the plural of side is sides; the

corner of any shape is called a vertex; the plural of vertex is vertices.

Square: 4 sides and 4 vertices; all its sides are equal in length.

Triangle: 3 sides and 3 vertices.

Key Facts: All sides of a square are equal in length (book misprints 'sqaure'). Opposite sides of a rectangle are equal in length. The sides of a triangle may or may not be equal in length.

Try Yourself: A circle has 0 sides and 0 vertices (it has one curved boundary and no corners).

Exercise 1 — 2-D Shapes

Square: 'My name is square. I have 4 sides and 4 vertices. My all sides are equal in length.'

Triangle: 'I have 3 sides and 3 vertices. Who am I? Triangle.'

Rectangle: 'I am a rectangle. I have 4 sides and 4 vertices.'

Try Yourself (oval): 'My name is oval. I have 0 sides and 0 vertices.'

Unit 6: Geometry — Lines, Patterns, 3-D Shapes

Straight and Curved Lines

A thread pulled tightly in both hands is like a straight line (straight path to the house). A thread held loose is like a curved line (curved path to the house). Straight lines can be horizontal, vertical or slanting; curved lines bend in many ways.

Exercise 2 — Straight and Curved Lines

Match: Straight Lines = the slanting, vertical and horizontal straight lines; Curved Lines = the wavy, S-shaped and arc lines.

Count straight and curved lines: (a) Curved diamond shape: Straight lines 2,

Curved lines 2. (b) Vase: Straight lines 2 (top rim and base), Curved lines 2 (the two sides). (c) Dome with two small side domes: Straight lines 1 (the base line), Curved lines 3 (big dome arc + 2 small arcs).

Count again: (a) Square: Straight lines 4, Curved lines 0. (b) Semi-circle: Straight lines 1 (base), Curved lines 1 (arc). (c) Triangle: Straight lines 3, Curved lines 0.

Drawing a straight line: Use a ruler. We measure and draw a straight line with the help of the ruler.

Measuring and Drawing Straight Lines

The given straight line measures from 0 to 6 on the ruler: its length is 6 cm.

To draw an 8 cm line: hold the ruler firmly on the paper and draw a straight line from 0 cm to 8 cm with a pencil, then remove the ruler.

Key Fact: To measure and draw a straight line using the ruler, we start from 0 cm.

Exercise 3 — Measure and Draw Lines

Measure the printed lines with a ruler (lengths as printed in the PDF at 100% scale, measured from the render — print-scale dependent): (a) 5 cm. (b) 7 cm. (c) 12 cm.

Draw with a ruler: (a) 4 cm long straight line. (b) 6 cm long straight line. (c) 10 cm long straight line. Method: place the ruler's 0 at the start point and draw to the required mark.

Patterns

Pattern of change in shape: big yellow triangle, small yellow square, repeating.

Pattern of change in size: big blue quarter-circle, small blue quarter-circle, repeating.

Pattern of change in direction: green triangle pointing right, green triangle pointing up, repeating.

Pattern of change in shape and colour: red semi-circle, brown rectangle, repeating.

Pattern of change in size and colour: small orange rectangle (2 squares), big purple rectangle (6 squares), repeating.

Exercise 4 — Complete the Grid Patterns

Row 1 (blue): horizontal bar of 3 squares, then vertical bar of 2 squares, repeating — next: a horizontal 3-square bar, then a vertical 2-square bar.

Row 2 (red): red triangle (2 squares wide), red square, repeating — next: a red triangle, then a red square.

Row 3 (purple): horizontal bar of 2 squares, then vertical bar of 2 squares, repeating — next: a horizontal 2-square bar, then a vertical 2-square bar.

Row 4 (green/yellow): green 2x2 block, yellow right-triangle (2x2), repeating — next: a green 2x2 block, then a yellow triangle.

Row 5 (red/blue): red triangle (2x2), blue 2x2 block, repeating — next: a red triangle, then a blue block.

Row 6 (brown/green): brown right-triangle, green right-triangle, repeating — next: a brown triangle, then a green triangle.

Row 7 (yellow/red): yellow trapezoid, red triangle, repeating — next: a yellow trapezoid, then a red triangle.

3-D Shapes

Cube (example: Rubik's cube). Cuboid (example: tissue box). Sphere

(example: football). Cone (example: ice-cream cone). Cylinder (example: glass).

Exercise 5 — Match Objects to 3-D Shapes

Eraser = cuboid. Stacking-rings toy = cone. Basketball = sphere. Alphabet block = cube. Drink can = cylinder.

I have learnt to: identify square, triangle, circle, semi-circle and quarter-circle; identify and differentiate between straight line and curved line; draw a straight line using a ruler; make and complete patterns according to shape, size and orientation; recognise the 3-D shapes: cube, cuboid, cylinder, cone and sphere.

Vocabulary: Semi-circle, Quarter-circle, Straight line, Curved line, Ruler, Patterns, Grid, 3-D Shapes, Cube, Cuboid, Cylinder, Sphere, Cone.

Review — Geometry

Multiple-choice answers: (i) 3 sides and 3 vertices = triangle (c). (ii) The quarter-circle shape = option (d). (iii) The cone shape = cone (b). (iv) A semi-circle has 1 curved line (b). (v) A circle has 0 sides (a).

Draw a rectangle, triangle and square using the grid (drawing activity).

Semi-circles and quarter-circles in the figure: Semi-circles 2 (middle dome + middle bowl); Quarter-circles 4 (left two + right two).

Robot drawing: Circle 4 (given: head + top + 2 ears). Square 2 (the two eyes). Rectangle 9 (mouth + neck + body + 2 arms + 2 legs + 2 feet). Triangle 6 (hat + nose + 2 hands + 2 foot-tops).

Draw a straight line of your own choice (drawing activity). Draw two patterns of your own choice and colour them (student activity).

Match 3-D shapes to names: red cuboid = Cuboid; blue sphere = Sphere;

orange cube = Cube; green cylinder = Cylinder; purple cone = Cone.

The book ends at page 151 with the Qaumi Tarana (national anthem) — no exercise.